INCH-POUND

MIL-T-49311 (CR)

5 October 1989

MILITARY SPECIFICATION

TEST FACILITIES SET AN/TPM-24()(V)4

This specification is approved for use by the Communications-Electronics Command, Fort Monmouth, Department of the Army, and is available for use by all departments and agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers Test Facilities Set AN/TPM-24()(V)4 The Test Facilities Set AN/TPM-24()(V)4 provides interconnection between Signal Processor CP-1187()TPX-46(V)7 and Radar Test Set AN/TPM-25A and consists of the following:

Hybrid Attenuator Assembly CN-1322()/TPM-24(V)
Interface Components

2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards The following specifications and standards form a part of this specification to the extent specified herein Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation

SPECIFICATIONS

MILITARY

MIL-P-11268	-Parts, Materials, and Processes Electronic Equipment
MIL-M-13231	-Marking of Electronic Items
MIL-F-14072	-Finishes for Ground Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN:AMSEL-ED-TO, Fort Monmouth, New Jersy, 07703-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC - N/A FSC 6625

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MIL-P-15024/6	-Plate, Identification-Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment
MIL-A-27434/24	Adapter, Connector, Coaxial, Radio Frequency, Between Series UG-1034()/U
MIL-A-55339/7	Adapter, Connector, Coaxial, Radio Frequency, (Within Series N), Class 2, Straight Plug

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	-Classification of Visual and Mechanical Defects for Equipment, Electronics, Wired and Other Devices
MIL-STD-454	-Standard General Requirements for Electronics Equipment
MIL-STD-461	-Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	-Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	-Environmental Test Methods
MIL-STD-1472	-Human Engineering Design Criteria for Military Systems Equipments and Facilities

*(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2 1 2 Other government documents, drawings, and publications The following other government documents, drawings, and publications form a part of this specification to the extent specified herein Unless otherwise, specified, the issues shall be those in effect on the date of the solicitation

DRAWINGS

SC-D-72336	Adapter, Connector UG-491B/U
SM-B-604411	Grounding Strap Assembly
SM-C-687974	Adapter, Connector UG-1896/TPM-24(V)
SM-C-687975	Adapter, Connector UG-1897/TPM-24(V)
SM-B-787960	Interface Unit, Dummy Connector J-3984/TPM-24(V)
SM-C-604306	Dummy Load, Electrical, Coaxial DA-636/U

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SM-C-604317-1	Dummy Load, Electrical DA-634/U
SM-C-604317-2	Dummy Load, Electrical DA-635/U
SM-C-687972	Dummy Load, Electrical DA-558/TPM-24(V)
SM-C-687977	Dummy Load, Electrical DA-559/TPM-24(V)
SM-C-787928	Extractor Circuit Board MX-10412/TPM-24(V)
SM-D-604227	Hybrid Attenuator Assembly CN-1322A/TPM-24(V)
SM-D-604230	Cable Assembly, Radio Frequency CG-3754/TPM-24(V)
SM-D-604236	Cable Assembly, Special Purpose, Electrical CX-12916/TPM-24(V)
SM-D-604244	Extender Card, Electronic Test MX-9356/TPM-24(V)
SM-D-604256	Cable Assembly, Radio Frequency CG-3750/TPM-24(V)
SM-D-986522	Cable Assembly, Radio Frequency CG /TPM-24(V)
SM-D-787808	Cable, DC Power
SM-D-787809	Cable, AC Power
SM-D-986520	Case, Transit CY-8015/TPM-24(V)
SM-D-986521	Case, Accessory CY-8014/TPM-24(V)
SM-B-986527	Tuning Tool
SM-C-986528	Wrench 5/16
SM-C-986529	Wrench 15/64
SM-C-986526	Adapter, Connector MX-10408/TPM-24(V)
SM-C-986525	Converter, Radio Frequency CV-3747/TPM-24(V)
SM-B-986523	Adapter, Connector MX-10411/TPM-24(V)
SM-B-986532	Adapter Connector MX-10409/TPM-24(V)
SM-D-986530	Cable Assembly
SM-D-986531	Cable Assembly
SM-B-986524	Adapter, Connector MX-10410/TPM-24(V)
SM-D-986535	Cable Assembly RF
SM-D-986536	Cable Assembly RF
SM-A-986540	-Test Procedure Group A
DL-SM-A-986541	-Test Facilities Set AN/TPM-24(V)

(Copies of drawings required in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2 2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specificationshall take precedence Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3 1 First article When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

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3 2 Construction. The test facilities set shall be constructed in accordance with the requirements of this specification and of DL-SM-B-986541 The Hybrid Attenuator Assembly CN-1322()/TPM-24(V)4 and interface components drawings or specifications are listed as follows·

Item	Quantity	Drawing or specification
Hybrid Attenuator Assembly CN-1322()/TPM-24(V) with Calibration Charts	1 each	SM-D-604227
Interface Unit, Dummy Connector J-3984/TPM-24(V)	1 each	SM-B-787960
Extractor, Circuit Board MX-10412/TPM-24(V)	1 each	SM-C-787928
Extender Card, Electronic Test MX-9356()/TPM-24(V)	2 each	SM-D-604244
Cable, DC Power	1 each	SM-D-787808
Cable, AC Power	1 each	SM-D-787809
Cable Assembly, Radio frequency CG-3750()/TPM-24(V) (6 ft 0 in)	2 each	SM-D-604256
Cable Assembly, Radio Frequency CG- ()/TPM-24(V) (8 ft 0 in)	1 each	SM-D-986522
Cable Assembly, Radio frequency CG-3754()/TPM-24(V) (3 ft 0 in)	5 each	SM-D-604230
Cable Assembly, Special Purpose Electrical CX-12916()/TPM-24(V) (2 ft 0 in.)	3 each	SM-D-604236
Cable Assembly, Special Purpose	1 each	SM-D-986530
Cable Assembly, Special Purpose	1 each	SM-D-986531
Dummy Load, Electrical DA-559()/TPM-24(V)	2 each	SM-C-687977
Cable Assembly RF	1 each	SM-D-986535
Cable Assembly RF	1 each	SM-D-986536

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Dummy Load, Electrical DA-634()/U	1 each	SM-C-604317-1
Dummy Load, Electrical DA-634()/U	1 each	SM-C-604317-2
Dummy Load, Electrical DA-636()/U	1 each	SM-C-604306
Adapter, Connector UG-29()/U	2 each	MIL-A-55339/7
Adapter, Connector UG-491()/U	2 each	SC-D-72338
Adapter, Connector UG-57B/U	1 each	
Adapter, Connector UG-1034()/U	1 each	MIL-A-27434/27
Adapter, Connector UG-1896()/TPM-24(V)	4 each	SM-C-687974
Adapter, Connector UG-1897()/TPM-24(V)	2 each	SM-C-687975
Grounding Strap Assembly	2 each	SM-B-604411
Detector RF, HP8473B	1 each	
Adapter, Connector MX-10411/TPM-24(V)	1 each	SM-C-986523
Adapter, Connector MX-10410/TPM-24(V)	1 each	SM-C-986524
Adapter, Connector MX-10408/TPM-24(V)	1 each	SM-C-986526
Tuning Tool	1 each	SM-B-986527
Case Accessory CY-8014/TPM-24(V)	1 each	SM-D-986521
Case Transit CY-8015/TPM-24(V)	1 each	SM-D-986520
Wrench 5/16	1 each	SM-C-986528
Wrench 15/64	1 each	SM-C-986529
Adapter, Connector MX-10409/TPM-24(V)	1 each	SM-C-986532

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3 3 Parts, materials and processes In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply Selection of microcircuits shall be in accordance with MIL-STD-1562 (see 4 4)

3 4 Recycled, virgin and reclaimed materials It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended used of the manufactured item

3 5 Finish All surfaces of the Test Facilities Set AN/TPM-24(V)4 requiring a protective coating shall be finished in accordance with MIL-F-14072 and the equipment drawings specified herein (see 4 4)

3 6 Nameplate and marking

3 6 1 General Marking of Test Facilities Set AN/TPM-24(V)4 shall be in accordance with MIL-M-13231 (see 4 4)

3 6 2 Visibility Whenever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment

3 6 3 Nameplate Each nomenclatured item shall have a nameplate conforming to MIL-P-15024/6, Type H or B (see 4 4)

3 6 4 Serial Numbers Each Hybrid Attenuator Assembly CN-1322()/TPM-24(V)4 shall be serial numbered (see 4 4)

3 7 Systems safety engineering

3 7 1 Personnel hazards Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 and paragraph 5 13 of MIL-STD-1472 Compliance with these requirements will be verified through a visual inspection (see 4 12)

3 7 2 Corner and edge rounding All exposed corners and edges shall be rounded to minimum of 75mm (0 3 in) radius Sharp edges and corners that present a personnel or safety hazard or potential damage to equipment during use shall be suitably protected or rounded to a minimum radius of 13mm (0 5 in)

3 7.3 Radioactive materials Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses)

3 8 Interchangeability Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (see 4.8).

3.9 Weight. The Hybrid Attenuator Assembly CN-1322/TPM-24(V)4 weight shall not exceed 12 5 lbs (5 7kg). The Case, Transit CY-8015/TPM-24(V)4 weight shall not exceed 40 lbs (18.14kg) (see 4.7).

3 10 Dimension. The external dimension of Hybrid Attenuator Assembly

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CN-1322/TPM-24(V)4 are: Width 13 Inches, (32 02cm) Height 894 Inches (22 70cm); Depth: 7 Inches (17 78cm) (See 4 6)

3.11 Performance requirements The Test Facilities Set AN/TPM-24(V)4 shall meet the specified performance requirements of SM-A-986540 (see 4 9)

3 12 Service conditions The Test Facilities Set AN/TPM-24()(V)4 shall meet the following service conditions (see 4 10)

3.12 1 Altitude The Test Facilities Set AN/TPM-24(V)4 shall meet specified performance, without degradation, at altitude up to and including 10,000 feet (3048m) above sea level, and shall withstand air transportation at altitude up to 40,000 feet (12,192m) above sea level (see 4 10 1)

3.12.2 High temperature

3 12 2 2 Operating The Test Facilities Set shall be operable without deterioration damage or degradation in specified performance at ambient air temperature as high as +131°F (55°C) (see 4 10 2)

3 12 2.2 Nonoperating The Test Facilities Set shall withstand exposure to ambient air temperature as high as +160°F (71°C) (see 4 10 2)

3 12 3 Low temperature

3 12 3 1 Operating The Test Facilities Set shall be operable without degradation or deterioration in specified performance at ambient air temperature as low as -32°F (-35 5°C) (see 4 10 3)

3 12 3 2 Nonoperating The Test Facilities Set shall withstand exposure to ambient air temperature as low as -70°F (-57°C) (see 4 10 3)

3 12 4 Humidity The Test facilities Set shall be operable without degradation in specified performance and shall sustain no physical damage, during and after prolonged exposure to extreme humidity levels as encountered in tropical areas (see 4 10 4)

3.12.5 Salt fog After the salt fog test of 4 10 5, the Test Facilities Set shall operate without degradation in specified performance and show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surface exfoliation There shall also be no clogging or binding of moving mechanical parts

3 12 6 Fungus After the fungus test of 4 10 6, the Test Facilities Set shall be visually examined The Test Facilities Set shall show no more than sparse microbial growth with restricted tabular growth development in an area 10 percent or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients In addition, the Test Facilities Set shall operate without degradation in specified

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performance after exposure to the test conditions of 4 10.6 Materials used in the manufacture, fabrication, construction and treatment of the Test Facilities Set shall comply with Requirement 4 of MIL-STD-454

3 12 7 Vibration The Test Facilities Set shall operate without degradation in specified performance and shall not suffer any mechanical damage after being subjected to the vibration test specified in 4 10 7

3 12 8 Shock (bench handling) The Test Facilities Set shall operate without degradation in specified performance and shall suffer no mechanical damage after subjected to the shock test specified in 4 10 8

3 13 Electromagnetic interference The Test Facilities Set shall meet the requirements of MIL-STD-461A, Notice 4 (see 4.11)

3.14 Workmanship. The Test Facilities Set shall be manufactured and assembled in accordance with Requirement 8 of MIL-STD-454 and the applicable portions of the following paragraphs in MIL-P-11268 (see 4 12):

General requirements for plastic material and plastic parts
Wiring and cabling, including
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering the brazing
Cleaning
General Requirements for securing of parts
Cleaning prior to welding
Process for welding
Controls
Self-tapping screws
Cleaning after welding
Printed wiring assemblies

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4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4 1.1 Responsibility for compliance All items must meet all requirements of Sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products and supplies submitted to the government for acceptance comply with all requirements of the contract Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material

4 2 Classification of inspections The inspection requirements specified herein are classified as follows

a First article inspection (see 4 3), does not include packaging
b Inspections covered by subsidiary documents see 4 4)
c Quality conformance inspection of equipment before packaging (see 4 5)

4 3 First article Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor

4 3 1 First article units The contractor shall furnish three first article units of the Test Facilities Set for Groups A B and C testing

4 3 2 First article inspection The first article inspection shall consist of the inspections specified in Table I, and shall be performed in the order specified

4.3 3 First article data The first article test plan and test reports shall be as required in the contract

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials and processes	3.3
Finish	3.5
Nameplate and marking	3 6
Workmanship	3.14

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TABLE I First article inspection

	Inspections	Reqt para	Test para	Order of tests Unit 1	Unit 2	Unit 3
1	Inspections covered by subsidiary documents 1/	3 3 3 5 3 6	4 4	Inspection to be performed on all units		
2	Group A inspection 1/	See Table II		Tests to be performed on all units		
3.	Group B inspection 1/	See Table III		Tests to be performed on all units		
4	Group C inspection					
	Altitude	3 12 1	4.10.1		1	
	Low temperature	3 12 3	4 10 3	1		
	High temperature	3 12 2	4 10 2	2		
	Humidity	3 12 4	4 10 4	3		
	Salt fog	3 12 5	4 10 5		2	
	Vibration	3 12 7	4 10 7			2
	Shock (bench handling)	3 12 8	4 10 8			3
	Fungus	3 12 6	4 10 6			4
5	EMI	3.13	4 11			1

1/ The inspection 1 to 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements

4 5 Quality conformance inspection of equipment before packaging The contractor shall perform the inspections specified in 4 4, 4 5 1 through 4 5 3 This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements

4 5.1 Group A inspection Each unit on contract shall be inspected for conformance to the inspection in TABLE II Discrete lots shall be formed from units that pass this inspection Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105 Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105

4.5 1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the government

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TABLE II Group A inspection

Inspection	Requirement paragraph	Test paragraph
Visual and mechanical	3 14	4 12
System safety	3 7	4 12
Performance	3 11	4 9

4 5 2 Group B inspection Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and meet group A inspection This inspection shall conform to TABLE III

4 5 2 1 Group B sampling plans The group B sampling plans shall be inspection Level S-4

4 5 2 2 Order of inspection within group B Group B inspection shall be performed in an order satisfactory to the government

TABLE III Group B inspection

Inspection	Requirement paragraph	Test paragraph
Interchangeability	3 8	4 8
Weight	3 9	4 7
Dimension	3 10	4 6

4 5 3 Group C inspection Group C inspection shall be performed on inspection lots that have passed group A and group B inspections The inspection shall consist of the inspections specified in TABLE IV Samples shall be selected in accordance with 4 5 3 1

4 5 3 1 Sampling for group C inspection Three Test Facilities Set shall be selected at random, one sample for each subgroup I, II and III in TABLE IV from every 25 units or fraction thereof produced The first samples selected shall be at the start of the contract from the first quality conformance inspection lot The salt fog, electromagnetic interference and fungus tests shall be performed only on the first sample units selected for the first group C inspection

4.5.3.2 Group C failures Action required relative to group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection only

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TABLE IV Group C inspection

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
Low temperature	3 12 3	4 10.3
High temperature	3 12 2	4 10 2
Humidity	3 12 4	4 10 4
Subgroup 2		
Altitude	3 12 1	4 10 1
Salt Fog 1/	3 12 5	4 10 5
Subgroup 3		
EMI	3 13	4 11
Vibration	3 12 7	4 10 7
Shock	3 12.8	4 10 8
Fungus 1/	3 12 6	4 10 6

1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection

4 6 Dimensions The Hybrid Alternator Assembly, CN-1322/TPM-24()4 shall be measured to determine compliance with the requirement of 3 10

4 7 Weight The Test Facilities Set shall be weighed on a calibrated scale to determine compliance with the weight requirement of 3 6

4 8 Interchangeability The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the units Mechanical dimensions shall be measured to determine compliance with Requirement 7 of MIL-STD-454 (see 3 8)

4 9 Performance testing The Test Facilities Set shall be tested for compliance with specified performance requirements of SM-A-986540 Test Procedure Group A

4 10 Service conditions The service condition tests shall be performed as detailed below Each Test Facilities Set subjected to these tests shall have passed group A tests and shall be resubmitted to and pass these group A tests after completion of all service condition tests Where operation of the test item is required during any service condition test, the test item shall be required to pass an operational test of 4 9. Upon completion of each service condition test, the test item shall be required to pass the operational test

4 10 1 Altitude The Test Facilities Set shall be subjected to the test of Method 500.1 Procedure 1 of MIL-STD-810 The Test Facilities Set shall be tested at 40,000 ft (12,192m) non-operating for one hour. The operating altitude in Step 2 shall be 10,000 ft (3,048m) Inability of the Test Facilities Set to meet the requirements of 3 12 1 shall consititute failure of this test

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4 10 2 High temperature The Test Facilities Set shall be subjected to the test of MIL-STD-810, Method 501 1, Procedure II. In step 7, the temperature shall be 131°F (55°C). In step 8, initiate operation of the equipment and at the end of a 30 minute period or when temperature stabilization occurs (whichever occurs first) the equipment shall show no degradation in performance (see 3 12 2)

4 10 3 Low temperature The Test Facilities Set shall be subjected to the test of MIL-STD-810, Method 502 1, Procedure I The temperature in step 2 shall be maintained for a period of not less than two hours following stabilization of the test item Following the two hour stabilization period, raise the chamber temperature to the low operating temperature (step 4) of 31°F (0°C) and maintain until temperature stabilization of the test item is reached In step 5, initiate operation of the equipment and at the end of a 30 minute period or when temperature stabilization occurs (whichever occurs first) the equipment shall show no degradation in performance (see 3.12.3).

4 10 4 Humidity The Test Facilities Set shall be subject to the test MIL-STD-810, Method 507 1, Procedure III Prior to measurements, excess surface water may be removed from the equipment by wiping external surfaces only In step 6, operation of 4 9 and measurements shall be performed during the last - five hours of the last cycle (see 3 12 4)

4 10 5 Salt fog The Test Facilities Set (or selected samples of materials, finishes, coatings, hardware, vendor items, parts and cabling) shall be subjected to the test of Method 509 1, Procedure 1 of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blisterec or otherwise loosened finish or metal surface Approval of the procuring activity shall be obtained for all items selected for salt fog testing

4 10 6 Fungus testing The Test Facilities Set (or selected samples of materials, finishes, coatings, hardware, vendor items, parts and cabling) shall be subjected to the test of Method 508 1, Procedure 1 of MIL-STD-810 Test items selected shall be representative of materials, finishes, surface treatments, coatings, and manufacturing processes used in the final design and construction of the equipment Approval of the procuring activity shall be required for items selected for test Failure criteria shall be determined as follows: Following fungus testing, sample which show corroded, etched, stained, or otherwise degraded areas as a result of this test (as determined by a certified laboratory examination) which constitute more than 10 percent of the sample surface area shall be considered to have failed this test. Test items whose design construction make it difficult or impossible to determine the extent of degradation shall be disassembled as necessary and the above failure criteria shall then apply to each disassembled piece. If any one such piece is found to have more than 10 percent of its external or internal surface area degraded, then the entire assembled test item shall be considered to have failed the test.

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4 10 7 Vibration (secure cargo) The Test Facilities Set in its transit case shall be subjected to the text of Method 514.2, Procedure X, Curve AW of Figure 514 2-7 of MIL-STD-810. The cycling and sweep time shall be in accordance with TABLE 514 2-VII Inability of the Test Facilities Set to meet the requirements of 3 12 7 shall connstitute failure of this test

4 10 8 Shock (bench handling) The Test Facilities Set shall be subjected to the test of Method 516 2, Procedure V of MIL-STD-810 Inability of the Test Facilities Set to meet the requirements of 3 12 8 shall constitute failure of the test

4 11 Electromagnetic interference The Test Facilities Set shall be tested for compliance with the requirements of MIL-STD-461A, Notice 4, using the approved test methods of MIL-STD-462, Notice 3 The applicable requirements are·

CE02	CS02	RE02	RS02
CE04	CS06	RE02 1 (A)	RS03

The Test Facilities Set shall not malfunction, exhibit degraded performance or undesirable response when immersed in electric fields of the following intensity using the modulation characteristic specified·

Frequency range	Field intensity	Modulation characteristics
10KHz - 2MHz	1 volt/meter	AM, 50%, 1000Hz tone
2MHz - 30MHz	5 volt/meter	AM, 50%, 1000Hz tone
30MHz - 76MHz	10 volt/meter	FM, ± 8KHz deviation, 1000Hz tone
76MHz - 400MHz	10 volt/meter	AM, 50%, 1000Hz tone
400MHz - 2GHz	10 volt/meter	Pulse, 0 1 usec pulse width, 800 pps
2GHz - 12 4GHz	5 volt/meter	Pulse, 0 1 usec pulse width, 800 pps

4 11 1 Bonds and grounds Bonding for equipment current path returns, RF potentials and shock hazards shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection Surface preparation for bonds and ground shll be accomplished by removing all anodic film, grease, paint, lacquer, or other high resistance coatings from the immediate areas of contact The equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The DC resistance of bonds and grounds shall not exceed 2 5 milliohms The bonding impedance shall be recorded and included in the EMI Test Report

4 11.2 Cables and connectors Shields cables, when required, shall have a shield weave providing not less than ninety percent coverage The shield shall be constructed of braided, tinned copper wire The use of shielded cables may be kept to a minimum by the use of alternative suppression techniques such as pin filters All panel mating surfaces shall have a RF conductive finish and provide positive bonding and grounding conductors and equipment ground plane.The DC bonding resistance for all connectors and panel mounted components shall not exceed 2 5 milliohms

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4 12 Visual and mechanical. The Test Facilities Set shall be examined for the defects listed in MIL-STD-252 In addition, the Test Facilities Set shall be visually inspected to determined conformance to the requirements of 3 7

5 PACKAGING

5 1 Packaging requirements The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6 2)

6 NOTES (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

6 1 Intended use The test Facilities Set AN/TPM-24(V)4 provides interconnection between Signal Processor CP-1187()TPX-46(V)7 and Radar Test Set AN/TPM-25A and consists of the following·

The Hybrid Attenuator CN-1322A/TPM-24(V) is provided for use in performing RF measurements in RF sum and difference channels The cable assemblies are provided for power connections, test equipment connections and for extending the modules of Signal Processor

Minor components are used in performing specific maintenance procedures on Test Facilities Set

6 2 Acquisition requirements Acquisition documents must specify the following·

a Title, number, and date of the specification

b Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2 1)

c Levels of preservation and packing (see Section 5).

d Marking and shipping of samples

e Place of final inspection.

f. Number of first articles to be submitted for approval (see 3.1).

g. When first article packaging inspection reports require acquisition activity approval prior to production unit packing

h Nameplate data

6.3 Details to be specified by the procuring activity.

6.4 Acceptability quality level (AQL) The following AQL's are recommended for use by the procuring activity in connection with the inspections outlined below·

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a. Group A inspection

Inspection	Major	Minor
Visual and mechanical	1%	4%
System safety	1%	4%
Performance	1%	1/

1/ All electrical defects are major

b Group B Inspection

Inspection	ALQ
Interchangeable	6.5%
Weight	6 5%
Dimension	6 5%

6 5 Nomenclature The parentheses in the nomenclature will be deleted andreplaced by a letter identifying the particular design, for example,AN/TPM-24(V)4 The contractor shall apply for nomenclature in accordance with applicable clause in the contract (see 1 1)

6 6 Subject terms (key word) list

Interconnection cables
Minor maintenance components
Signal processor CP-1187()TPX-46(V)7
Radar Test Set AN/TPM-25A
Hybrid Attenuator CN-1322A/TPM-24(V)
Test Facilities Set AN/TPM-24()(V)4

Custodian
Army - CR

Preparing activity
Army - CR
Project 6625-A806

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-49311 (CR)	TEST FACILITIES SET AN/TPM-24()(V)4

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street City State ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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